

End-to-End Process · Plan-to-Produce (P2Pr)

Process Specification

Step-level specification: People · Process · Data / CRUD · Technology · PCF 4.1 + 4.3 + 4.4 · Functions: Engineering · Materials · Manufacturing · Supplier · Handoff to P2P at PCF 4.2

Step · PCF	Pattern	M3 H%	People	Process	Data · CRUD	Technology
Engineering · PCF 4.3 — BOMs, Routings, and Product Specifications						
Develop Bill of Materials 4.3.1	DOCUMENT	40%	AI generates draft BOM from design specs. Engineer reviews, validates, and approves.	Define components, quantities, and assembly hierarchy. Manage multi-level BOM structure.	C Bill of Materials · PLM / ERP R Product Design · CAD / PLM R Item Master · ERP MM	PLM — Teamcenter / Windchill ERP MM — SAP MM AI BOM generation
Define Production Routings 4.3.1	KNOWLEDGE	65%	Process Engineer — AI suggests routing based on similar products. Human validates.	Define work centers, operations, standard times, and production sequence per BOM.	C Production Routing · ERP PP R Bill of Materials · PLM / ERP R Work Center · ERP PP	ERP PP — SAP PP / Oracle MFG PLM AI routing optimization
Manage Engineering Change Orders 4.3.1	DECISION	80%	Engineering Manager — ECO approval is irreducibly human. AI impact-assesses change.	Receive, evaluate, and approve/reject engineering changes. Update BOM and routing. Notify Manufacturing.	U Bill of Materials · PLM / ERP U Production Routing · ERP PP C Change Order · PLM	PLM — ECO workflow ERP MM / PP AI change impact analysis
Materials · PCF 4.1 + 4.4 — Supply Chain Planning, Inventory, and Logistics						
Develop Supply Chain Strategy 4.1.1	KNOWLEDGE	65%	Supply Chain Manager — AI models network scenarios. Human sets strategic direction.	Define sourcing strategy, supply network design, and make-vs-buy decisions.	C Supply Chain Plan · SCM Platform R Demand Data · ERP / BI R Supplier Master · ERP MM	SCM platform — SAP IBP / Kinaxis AI network optimization BI platform
Develop Demand Forecast 4.1.2	KNOWLEDGE	65%	Materials team — AI generates statistical forecast. Human adjusts for market intelligence.	Build demand plan from sales history, pipeline, and market data. Align with Sales forecast.	C Demand Forecast · SCM Platform R Sales History · ERP SD R Sales Forecast · CRM / SCM	SCM platform — SAP IBP / Kinaxis AI demand sensing CRM integration
Develop Inventory Plan 4.1.3	DECISION	80%	Materials Manager — safety stock and reorder decisions carry financial authority.	Set safety stock levels, reorder points, and min/max parameters by item and location.	U Item Master · ERP MM R Demand Forecast · SCM Platform R Supplier Lead Time · ERP MM	ERP MM — SAP MRP SCM platform AI inventory optimization
Develop Production Plan 4.1.4	DECISION	80%	Materials / Production Manager — production plan approval carries capacity commitment.	Convert demand forecast into production requirements. Balance capacity and inventory targets.	C Production Plan · ERP PP R Demand Forecast · SCM Platform R Capacity Plan · ERP PP	ERP PP — SAP PP / MRP II SCM platform — Kinaxis / SAP IBP AI capacity optimization
Manage Supply Chain Risk 4.1.5	KNOWLEDGE	65%	Materials team — AI monitors signals. Human evaluates and activates risk response.	Monitor supplier risk, geopolitical events, and supply disruptions. Maintain contingency plans.	R Supplier Master · ERP MM R Supply Chain Plan · SCM Platform C Risk Record · Risk Platform	Risk platform SCM platform AI risk monitoring — supply chain intelligence
Receive and Store Materials 4.4.1	TRANSACTION	35%	Warehouse staff — AI directs put-away. Human executes	Receive inbound materials from Supplier. Verify	C Goods Receipt · ERP WM	ERP WM — SAP WM / EWM

			physical receipt and storage.	against PO. Put away to storage location.	R Purchase Order · ERP MM U Inventory · ERP WM	Barcode / RFID scanning AI put-away optimization
Manage Inventory 4.4.2	TRANSACTION	35%	AI — automated inventory transactions. Materials team manages exceptions and cycle counts.	Maintain inventory accuracy. Process stock movements, transfers, and adjustments.	U Inventory · ERP WM R Item Master · ERP MM C Stock Movement · ERP WM	ERP WM — SAP WM / EWM AI inventory optimization Cycle count automation
Planning Exception 4.1.2	EXCEPTION	82%	Materials Manager — human resolves supply/demand imbalance. AI flags and quantifies.	Investigate forecast error, supply disruption, or capacity constraint. Adjust plan or escalate.	U Production Plan · ERP PP U Demand Forecast · SCM Platform R Supplier Master · ERP MM	SCM platform exception workflow ERP PP AI exception alerting
Manufacturing · PCF 4.3 — Production Scheduling, Execution, and Quality						
Develop Production Schedule 4.3.2	KNOWLEDGE	65%	Production Scheduler — AI optimizes sequence. Human approves and manages constraints.	Sequence work orders across work centers. Balance load and minimize changeover. Publish schedule.	C Production Schedule · ERP PP / MES R Production Plan · ERP PP R Work Center · ERP PP	ERP PP / MES — SAP MES / Opcenter AI scheduling optimization Finite capacity scheduling
Manage Work Orders 4.3.3	TRANSACTION	35%	AI — automated work order creation and release. Production Supervisor manages exceptions.	Create, release, and track work orders against production schedule. Confirm materials availability.	C Work Order · ERP PP / MES R Production Schedule · ERP PP R Bill of Materials · PLM / ERP R Inventory · ERP WM	ERP PP / MES AI work order optimization MES — real-time tracking
Execute Production 4.3.4	TRANSACTION	35%	Production operators — AI monitors and guides. Human executes physical assembly and test.	Execute production operations per routing. Record time, quantity, and quality data at each step.	U Work Order · ERP PP / MES R Production Routing · ERP PP U Inventory · ERP WM	MES — SAP MES / Rockwell IoT / shop floor data collection AI process monitoring
Perform Quality Inspection 4.3.5	DECISION	80%	Quality Inspector — accept/reject decision is irreducibly human. AI assists with defect detection.	Inspect finished or in-process goods against specification. Record results. Accept or reject.	C Inspection Lot · ERP QM R Work Order · ERP PP / MES U Work Order · ERP PP / MES	ERP QM — SAP QM AI vision inspection MES quality module
Manage Non-Conformance 4.3.6	EXCEPTION	82%	Quality Engineer + Manufacturing Manager — human disposition decision. AI logs and routes.	Investigate defect root cause. Disposition: rework, scrap, or use-as-is. Raise corrective action.	C Non-Conformance Record · ERP QM U Work Order · ERP PP / MES U Inventory · ERP WM	ERP QM — NCR workflow CAPA system AI root cause analysis
Record Production Output 4.3.7	TRANSACTION	35%	AI — automated goods receipt from production. Operator confirms physical completion.	Post production output to inventory. Update work order status. Trigger inventory update to Materials.	U Work Order · ERP PP / MES U Inventory · ERP WM C Goods Receipt · ERP PP	ERP PP — SAP MFGR GR MES AI automated posting
Production Exception 4.3.4	EXCEPTION	82%	Production Manager — human resolves. AI flags capacity, material, or quality breach.	Investigate production stoppage, yield loss, or schedule miss. Escalate or recover.	U Work Order · ERP PP / MES U Production Schedule · ERP PP R Non-Conformance Record · ERP QM	MES exception workflow ERP PP AI anomaly detection

CRUD Data Matrix · P2Pr End-to-End Process

C = Create · R = Read · U = Update · D = Delete · · = No operation at this step

Engineering · PCF 4.3 — BOMs, Routings, and Product Specifications

Data Entity	System of Record	4.3.1 (1)	4.3.1 (2)	4.3.1 (3)
Bill of Materials	PLM / ERP	C	R	U
Production Routing	ERP PP	·	C	U
Product Design	CAD / PLM	R	·	·
Item Master	ERP MM	R	·	·
Work Center	ERP PP	·	R	·
Change Order	PLM	·	·	C

(1) 4.3.1 Develop Bill of Materials · (2) 4.3.1 Define Production Routings · (3) 4.3.1 Manage Engineering Change Orders

Materials · PCF 4.1 + 4.4 — Supply Chain Planning, Inventory, and Logistics

Data Entity	System of Record	4.1.1 (1)	4.1.2 (2)	4.1.3 (3)	4.1.4 (4)	4.1.5 (5)	4.4.1 (6)	4.1.2 (7)
Supply Chain Plan	SCM Platform	C	·	·	·	·	·	U
Demand Forecast	SCM Platform	·	C	·	·	·	·	U
Production Plan	ERP PP	·	·	·	C	·	·	U
Item Master	ERP MM	·	·	U	·	·	·	·
Inventory	ERP WM	·	·	·	·	·	U	U
Goods Receipt	ERP WM	·	·	·	·	·	C	·
Supplier Master	ERP MM	R	·	·	·	R	·	R
Risk Record	Risk Platform	·	·	·	·	C	·	·

(1) 4.1.1 Develop SC Strategy · (2) 4.1.2 Develop Demand Forecast · (3) 4.1.3 Develop Inventory Plan · (4) 4.1.4 Develop Production Plan · (5) 4.1.5 Manage SC Risk · (6) 4.4.1 Receive and Store Materials · (7) 4.1.2 Planning Exception

Manufacturing · PCF 4.3 — Production Scheduling, Execution, and Quality

Data Entity	System of Record	4.3.2 (1)	4.3.3 (2)	4.3.4 (3)	4.3.5 (4)	4.3.6 (5)	4.3.7 (6)	4.3.4 (7)
Production Schedule	ERP PP / MES	C	R	·	·	·	·	U
Work Order	ERP PP / MES	·	C	U	·	U	U	U
Inventory	ERP WM	·	·	U	·	U	U	·
Inspection Lot	ERP QM	·	·	·	C	·	·	·
Non-Conformance Record	ERP QM	·	·	·	·	C	·	R
Goods Receipt	ERP PP	·	·	·	·	·	C	·
Production Routing	ERP PP	·	·	R	·	·	·	·

(1) 4.3.2 Develop Production Schedule · (2) 4.3.3 Manage Work Orders · (3) 4.3.4 Execute Production · (4) 4.3.5 Perform Quality Inspection · (5) 4.3.6 Manage Non-Conformance · (6) 4.3.7 Record Production Output · (7) 4.3.4 Production Exception